

HADI AHMAD

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EDUCATION

UNIVERSITY OF MINNESOTA – TWIN CITIES

Bachelor of Science in Computer Science and Electrical Engineering

Minneapolis, MN

Expected May 2028

University Honors Program | **Dean's List** | **3.7 Technical GPA**

Relevant Coursework: Database Systems, Operating Systems, Parallel Programming, Data Modeling, Machine Learning, Computer Architecture, Microcontrollers

Awards: Presidential Scholarship, Iron Range Scholarship, Dakota Electric Association Scholarship

Involvement: Minnesota Nanofabrication Club (Ultrasonic Cleaner Lead)

PROJECTS

ROOMINATE | *React Native, TypeScript, Python, PostgreSQL* | roominate.me

72 hours in September 2026

Campus Study-Room Finder (0-1 Startup Hackathon)

- Built a cross-platform **iOS/Android/web** app in **React Native** and **TypeScript** surfacing free study rooms across **574** rooms in **73** UMN buildings, ranked by walking time and how long each room stays free; localized the UI into **11** languages.
- Validated the availability engine, Python scraper, and database schema with **140+ automated tests (Jest, unittest, SQL)**.
- Engineered a **Python ETL** pipeline filtering **1,100+** 25Live campus spaces to student-accessible rooms across 4 access categories, publishing **~8,500** reservations to **Supabase/PostgreSQL** every 30 min via scheduled **GitHub Actions** workflows.
- Moved availability computation on-device, eliminating client calls to a feed returning **8 MB** of XML in **~10 s** per query.
- Designed a data-trust heuristic flagging rooms w/ bookings under **10%** of the campus median as unmaintained calendars.

NEURAL FREQUENCY INTERFACE | *Verilog, Vivado, Python, KiCad, SKiDL* | **Personal Project**

June 2026 – Present

Real-Time EEG Neurofeedback Interface System

- Built a real-time **FPGA DSP** pipeline in **Verilog**: 32 parallel **FIR** filters across 8 channels and an LMS adaptive artifact filter.
- Engineered **AES-128-CCM** authenticated UART/BLE telemetry link @ **230,400 baud** w/ hardware flow control: 135B bundled frames every 8ms for a 0.5ms macro scheduling **guardband** over BLE's 7.5ms floor, absorbing radio-side lag.
- Reduced **BLE** power draw by **~88.4%** with 30B 50Hz packet protocol; extending total battery life by estimated **~12%**.

PLANUMN | *React, Next.js, TypeScript, PostgreSQL, SQLite* | planu.mn

April 2025 – August 2025

Graduation Planner Web Application

- Launched a drag-and-drop, mobile-adaptive graduation planning web application used by **80+** UMN students pre-release.
- Integrated a **PostgreSQL** database schema via **Supabase**, with user authentication, RLS, and server-side permission checks.
- Built responsive UI components using **React**, **Next.js**, and **TypeScript**; server-side **SQLite** for fast course data fetching.
- Deployed a transcript import feature that parses unofficial UMN transcript PDFs to autofill each student's completed courses.

ULTRASONIC WAFER CLEANER | *KiCad, SPICE* | **Minnesota Nanofabrication Club**

Sep 2026 – Present

80 kHz Ultrasonic Cleaning System for Wafer Processing

- Lead the club's build of an **80 kHz** ultrasonic cleaner, with a custom **half-bridge driver** for three 60 W piezo transducers.
- Designing an **IPA vapor-drying** stage to displace water during wafer cassette withdrawal, eliminating residue formation.

PROFESSIONAL EXPERIENCE

ULTRASONIC CLEANER LEAD

Minnesota Nanofabrication Club • University of Minnesota

Minneapolis, MN

Sep 2026 – Present

UNDERGRADUATE TEACHING ASSISTANT

University of Minnesota • Department of Electrical & Computer Engineering

Minneapolis, MN

Aug 2026 – Present

RETAIL / FOOD SERVICE EXPERIENCE

McDonalds • Blaze Pizza • Planet Fitness • Target • Walmart

Lakeville, MN

Sep 2022 – May 2026

TECHNICAL SKILLS

LANGUAGES: Java, C/C++, Python, JavaScript/TypeScript, SQL, HTML/CSS, Swift, Verilog, MATLAB, R

WEB & CLOUD: React/React Native, Expo, Next.js, Node.js, PostgreSQL, AWS, Docker

TESTING & TOOLS: Jest, Vitest, Playwright, GitHub Actions, Git/GitHub, CUDA

DESIGN: Vivado, Simulink, KiCAD, Altium, LTSpice, MPLAB